
An Energy-Based View on the Manifold Hypothesis

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Abstract

1 A commonly adopted assumption in modern machine learning is the *manifold*
2 *hypothesis*, which claims that although data is embedded in a high-dimensional
3 ambient space, the probability mass concentrates near a much lower-dimensional
4 geometric structure. However, the real-world latent variables that images are drawn
5 from are continuous, not discrete, and therefore cannot be completely captured by
6 a finite manifold. Moreover, the assumption does not properly capture the union
7 of manifolds, stratified spaces, and manifolds with singularities and boundaries.
8 Drawing upon recent advances in energy-based models, we provide an energy-
9 based perspective on the real-world latents, describing an Occam’s razor style
10 analog for the manifold hypothesis, which is more amenable to recent literature
11 in the field. We theoretically prove that our construction recovers the manifold
12 hypothesis for coarse functions, and validate it with numerical examples.

13 1 Introduction

- 14 **1.1 Related Works**
- 15 **NERF** ??????????
- 16 **Energy-based models** ??????
- 17 **Compositional generation** ??????
- 18 **Occam's razor** ??

2 Problem Setting

Supervised Learning 101. Observing data $D = \{(x_i, y_i)\}_{i=1}^n$ we want to learn a deterministic function $f^* : \mathcal{X} \rightarrow \mathbb{R}$ such that $f^*(x_i) = y_i$ for all $i = 1, \dots, n$. Usually, we know the input space $\mathcal{X}$ via some base distribution ν_0 on $\mathcal{X}$, for instance, uniformly distribution over a compact set $\mathcal{X} \subset \mathbb{R}^d$. Moreover, we assume that smoother or more predictable the function f^* is, the better it generalizes to new data.

Notation. We refer to a metric space (X, d) as a Polish space if it is complete (every Cauchy sequence in (X, d) converges to a point in X) and separable (there exists a countable dense subset $D \subset X$). For instance, $\mathbb{R}^n$ with the Euclidean metric, separable Banach spaces, Hilbert spaces, and function spaces such as $L^2(M)$ or $H^s(M)$ (when M is a compact manifold) are all Polish spaces. Moreover, if X is Polish, then $(X, \mathcal{B}(X))$ enjoys strong regularity properties that make it a canonical measurable structure in probability theory, where $\mathcal{B}(X)$ is the Borel σ -algebra on X .

Definition 2.1 (Symmetric Dirichlet Space). Let $(X, \mathcal{F}, \mu)$ be a σ -finite measure space. A *Dirichlet form* on $L^2(\mu)$ is a pair $(\mathcal{E}, \mathcal{D}(\mathcal{E}))$ such that:

1. $\mathcal{D}(\mathcal{E}) \subset L^2(\mu)$ is a dense linear subspace;

2. $\mathcal{E} : \mathcal{D}(\mathcal{E}) \times \mathcal{D}(\mathcal{E}) \rightarrow \mathbb{R}$ is a symmetric bilinear form;

3. (Non-negativity)

$$\mathcal{E}(f, f) \geq 0 \quad \forall f \in \mathcal{D}(\mathcal{E});$$

4. (Closedness) $\mathcal{D}(\mathcal{E})$ is complete with respect to the norm

$$\|f\|_{\mathcal{E}}^2 := \|f\|_{L^2(\mu)}^2 + \mathcal{E}(f, f);$$

5. (Markov property) for every $f \in \mathcal{D}(\mathcal{E})$, the truncation $\tilde{f} := (0 \vee f) \wedge 1$ satisfies

$$\tilde{f} \in \mathcal{D}(\mathcal{E}) \quad \text{and} \quad \mathcal{E}(\tilde{f}, \tilde{f}) \leq \mathcal{E}(f, f).$$

The triple $(X, \mathcal{F}, \mu, \mathcal{E})$ is called a *Symmetric Dirichlet Space*.

The Dirichlet form encodes the geometry of the learning space via the energy functional: $\mathcal{E}(f, f)$ is the energy of the function f . Enforcing the Markov property allows us to define stochastic processes on the learning space: semi-group generated by $\mathcal{E}$ is a Markov semigroup.

The main idea of learning is that if $x_1 \approx x_2$ then $f(x_1) \approx f(x_2)$ for a function f . Dirichlet form enforces precisely this property. As it encodes the geometry of the learning space, it allows us to define a generalized Laplacian L on $L^2(\mu)$ and Sobolev classes of functions on X .

Definition 2.2 (Generalized Laplacian). Let $(X, \mathcal{F}, \mu, \mathcal{E})$ be a measure space endowed with a Dirichlet form. There exists a unique self-adjoint operator L on $L^2(\mu)$ such that

$$\mathcal{E}(f, g) = \langle f, Lg \rangle_{L^2(\mu)}, \quad \forall f \in \mathcal{D}(\mathcal{E}), g \in \mathcal{D}(L).$$

The operator L admits a spectral decomposition $L\phi_j = \lambda_j\phi_j$, with

$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \dots, \quad \langle \phi_j, \phi_k \rangle_{L^2(\mu)} = \delta_{jk}.$$

The Markov property of the Dirichlet form $\mathcal{E}$ implies that e^{tL} is a Markov semigroup. Hence, our distribution μ is a stable state of the system: given a base distribution ν_0 on $\mathcal{X}$, the distribution μ is the limit of the Markov process $e^{tL}\nu_0$ as $t \rightarrow \infty$.

Definition 2.3 (Effective dimension). The *effective dimension* of the learning space $(X, \mathcal{F}, \mu, \mathcal{E})$ is the quantity $d_{\text{eff}} \in (0, \infty)$ such that the eigenvalues of the corresponding Laplacian operator $L : L^2(\mu) \rightarrow L^2(\mu)$ satisfy

$$\lambda_j \asymp j^{2/d_{\text{eff}}} \quad \text{as } j \rightarrow \infty,$$

or equivalently, the eigenvalue counting function satisfies

$$N(\lambda) := \#\{j : \lambda_j \leq \lambda\} \asymp \lambda^{d_{\text{eff}}/2} \quad \text{as } \lambda \rightarrow \infty.$$

Or, more generally

$$\text{tr } e^{tL} \sim t^{-d_{\text{eff}}/2} \quad \text{as } t \rightarrow 0^+$$

56 **Definition 2.4** (Spectral Sobolev class). For smoothness $s > 0$ and radius $R > 0$, the *spectral*
 57 *Sobolev class* of order s is

$$\mathcal{H}_\mu^s(R) := \left\{ f \in L^2(\mu) : \|f\|_{\mathcal{H}^s}^2 := \sum_{j \geq 1} \lambda_j^s \widehat{f}_j^2 \leq R^2 \right\}, \quad \widehat{f}_j := \langle f, \phi_j \rangle_{L^2(\mu)}.$$

58 The condition $f \in \mathcal{H}_\mu^s(R)$ requires the spectral coefficients to decay as $|\widehat{f}_j| \leq R \lambda_j^{-s/2} \asymp R j^{-s/d_{\text{eff}}}$.
 59 higher smoothness s enforces faster decay and hence greater regularity with respect to the geometry
 60 encoded by $\mathcal{E}$.

61 **Definition 2.5** (Heat semigroup Besov space). Let $P_t = e^{-tL}$ be the Markov semigroup generated
 62 by L . For $s > 0$ and $1 \leq p, q \leq \infty$, define

$$\|f\|_{B_{p,q}^s(L)} = \|f\|_{L^p(\mu)} + \left(\int_0^1 \left(t^{-s/2} \|(I - P_t)f\|_{L^p(\mu)} \right)^q \frac{dt}{t} \right)^{1/q}.$$

63 Then $B_{p,q}^s(L)$ consists of all f for which this norm is finite.

64 **Definition 2.6** (Spectral wavelets associated to L). Let $\varphi_0 \in C_c^\infty([0, \infty))$ satisfy $\varphi_0(\lambda) = 1$ for
 65 $\lambda \in [0, 1]$ and $\text{supp } \varphi_0 \subset [0, 2]$. Define the dyadic wavelet kernel

$$\psi(\lambda) = \varphi_0(\lambda) - \varphi_0(2\lambda), \quad \psi_j(\lambda) = \psi(2^{-j}\lambda), \quad j \geq 1,$$

66 so that $\{\varphi_0, \psi_j\}_{j \geq 1}$ forms a partition of unity:

$$\varphi_0(\lambda) + \sum_{j=1}^{\infty} \psi_j(\lambda) = 1 \quad \text{for all } \lambda \geq 0.$$

67 The *wavelet blocks* are the operators

$$\Psi_0 = \varphi_0(L), \quad \Psi_j = \psi_j(L) = \psi(2^{-j}L), \quad j \geq 1,$$

68 defined via the spectral calculus of L , yielding the Littlewood–Paley decomposition

$$f = \Psi_0 f + \sum_{j=1}^{\infty} \Psi_j f.$$

69 If L has discrete spectrum $0 \leq \lambda_1 \leq \lambda_2 \leq \dots$ with eigenfunctions $\{e_k\}$, then

$$\Psi_j f = \sum_k \psi_j(\lambda_k) \langle f, e_k \rangle_{L^2(\mu)} e_k.$$

70 3 Manifold Hypothesis

71 A commonly adopted assumption in modern machine learning is the *manifold hypothesis*, which
 72 claims that although data is embedded in a high-dimensional ambient space, the probability mass
 73 concentrates near a much lower-dimensional geometric structure.

74 **Assumption 3.1** (Manifold Hypothesis). Let $\chi \subset \mathbb{R}^D$ be the ambient space and let $\mathcal{D}$ be a probability
 75 distribution on χ . There exists a d -dimensional manifold $\mathcal{M} \subset \chi$, with $d \ll D$, such that

$$\mathbb{P}_{x \sim \mathcal{D}}(\text{dist}(x, \mathcal{M}) \leq \varepsilon) \geq 1 - \delta,$$

76 for sufficiently small $\varepsilon, \delta > 0$.

77 In practice, data rarely concentrate on a single smooth manifold. Rather, they are better modeled as
 78 lying on *unions of manifolds*, or more generally on *stratified spaces* composed of pieces of varying
 79 intrinsic dimensions, potentially with singularities and boundaries.

80 Why small d is important?

- 81 • Generalization error depends on the intrinsic dimension d of the manifold.
- 82 • The number of samples n needed to learn the manifold increases with d .
- 83 • The complexity of the model increases with d .
- 84 • Can get a small embedding.

85 Shortcomings

- 86 • Doesn't properly capture union of manifolds.
- 87 • Doesn't properly capture stratified spaces.
- 88 • Doesn't properly capture manifolds with singularities and boundaries.
- 89 • What about infinite-dimensional data? Cause images come from infinite-dimensional
- 90 continuous functions.

91 [Emile says: would be nice to elaborate on the above, and to point to specific instances where
92 assuming it has some "catastrophic" consequences. that could go into a motivating examples section]

93 **Example: Vision Data** Each 2d image (or 3d scene or a point cloud) can be modeled $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$
94 as a function from coordinates to RGB (or XYZ) values. That's how NeRFs work:

95 **Neural Radiance Fields (NeRFs).** [Emile says: this actually looks very relevant. would be worth
96 doing a deeper dive into this] A *Neural Radiance Field* models a 3D scene as a continuous function

$$f_{\theta} : (\mathbf{x}, \mathbf{d}) \mapsto (\sigma, \mathbf{c}),$$

97 where

- 98 • $\mathbf{x} \in \mathbb{R}^3$ denotes a spatial location,
- 99 • $\mathbf{d} \in \mathbb{S}^2$ denotes a viewing direction,
- 100 • $\sigma \in \mathbb{R}_{\geq 0}$ is the volume density at $\mathbf{x}$,
- 101 • $\mathbf{c} \in \mathbb{R}^3$ is the emitted RGB color in direction $\mathbf{d}$.

102 The key idea is that the scene is represented neither as a mesh, nor as a voxel grid, nor as a point
103 cloud, but as a *continuous radiance field* parameterized by a neural network.

104 In a similar way, each 2d image can be modeled as a function from coordinates to RGB values:
105 $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and the embedding of this function would be the weights of the corresponding ReLU
106 network. The regular manifold hypothesis doesn't properly capture this structure that:

- 107 • In a union of manifolds
- 108 • Has some hierarchical structure.
- 109 • Has permutation-invariance with respect to weights in each layer.

110 Also, importantly, each image f is not a smooth function: its a combination of a "simple" in some
111 sense partition of the image into smooth enough regions.

112 4 Effective Dimension Hypothesis

113 Let us formulate the extension of the manifold hypothesis to this case. [Emile says: more generally,
114 need to define energy functions, why they're relevant to this problem, and what characteristics do we
115 want our manifold hypothesis to have?]

116 **Assumption 4.1** (Effective Dimension Hypothesis). All the observed data x_i come from a learning
117 space $(X, \mathcal{F}, \mu, \mathcal{E})$ with small effective dimension even if the ambient space $(X, \mathcal{F}, \nu_0, \mathcal{E})$ has large
118 effective dimension where ν_0 is the base distribution on $\mathcal{X}$:

$$d_{\text{eff}}(X, \mathcal{F}, \mu, \mathcal{E}) \ll d_{\text{eff}}(X, \mathcal{F}, \nu_0, \mathcal{E})$$

119 This assumption ensures two things: existence of the efficient coordinate system in the space $\mathcal{X}$ and
120 efficient learning rate for functions in Besov spaces. Moreover, this assumption of the underlying
121 learning space is strictly stronger than the manifold hypothesis and unifies multiple learning problems,
122 e.g. classical kernel methods, flow matching/diffusion models and transformers.

123 4.1 Generalization Error

124 hehe

125 **Proposition 4.2** (Bias-variance tradeoff). *Let $f^* \in \mathcal{H}_\mu^s(R)$ and suppose we observe $y_i = f^*(x_i) + \varepsilon_i$,
126 $i = 1, \dots, n$, with $x_i \sim \mu$ i.i.d. and ε_i i.i.d. sub-Gaussian with parameter σ^2 . Consider the truncated
127 spectral estimator*

$$\widehat{f}_k(x) = \sum_{j=1}^k \widehat{f}_j^{(n)} \phi_j(x), \quad \widehat{f}_j^{(n)} = \frac{1}{n} \sum_{i=1}^n y_i \phi_j(x_i).$$

128 Then the expected risk decomposes as

$$\mathbb{E}[\|\widehat{f}_k - f^*\|_{L^2(\mu)}^2] = \underbrace{\sum_{j>k} (\widehat{f}_j^*)^2}_{\text{bias}^2(k)} + \underbrace{\frac{\sigma^2 k}{n}}_{\text{variance}(k)}.$$

129 Under Effective Dimension Hypothesis (Assumption 4.1) ($\lambda_j \asymp j^{2/d_{\text{eff}}}$) and the Sobolev constraint
130 $\sum_j \lambda_j^s (\widehat{f}_j^*)^2 \leq R^2$:

131 **Bias bound.** The tail energy is controlled by the smoothness and dimension:

$$\text{bias}^2(k) = \sum_{j>k} (\widehat{f}_j^*)^2 \leq R^2 \lambda_{k+1}^{-s} \asymp R^2 k^{-2s/d_{\text{eff}}}.$$

132 **Variance bound.**

$$\text{variance}(k) = \frac{\sigma^2 k}{n}.$$

133 **Optimal truncation.** Balancing bias and variance:

$$k^{-2s/d_{\text{eff}}} \asymp \frac{k}{n} \implies k^* \asymp n^{d_{\text{eff}}/(2s+d_{\text{eff}})}.$$

134 **Minimax rate.** Substituting k^* into either term:

$$\boxed{\mathbb{E}[\|\widehat{f}_{k^*} - f^*\|_{L^2(\mu)}^2] \lesssim n^{-2s/(2s+d_{\text{eff}})}}.$$

135 This rate is minimax optimal over $\mathcal{H}_\mu^s(R)$.

136 **Remark 4.3** (Interpretation). The three quantities s , d_{eff} , and n interact as follows:

- 137 • **Smoothness s :** higher s means faster eigenvalue decay of the coefficients $\widehat{f}_j^*$, so the bias
138 decreases faster with k and fewer modes suffice.
- 139 • **Effective dimension d_{eff} :** larger d_{eff} means eigenvalues $\lambda_j \asymp j^{2/d_{\text{eff}}}$ grow more slowly, so
140 more modes carry significant energy and the variance cost of including them is higher. This
141 is the curse of dimensionality in d_{eff} , not in the ambient dimension.
- 142 • **Optimal number of modes k^* :** this is the resolution at which you can reliably estimate the
143 function given n samples. Modes $j \leq k^*$ are estimated; modes $j > k^*$ are discarded as
144 noise.

145 **Theorem 4.4** (Minimax rate over Besov balls ?). *[TODO: Replace $\mathbb{R}^d$ by $\mathcal{X}$] Let $f \in \mathcal{B}_{p,q}^s(R) \subset$
146 $L^r(\mathbb{R}^d)$ with $s > d(1/p - 1/r)_+$. Then the minimax L^r estimation rate from n noisy samples is*

$$\inf_{\widehat{f}_n} \sup_{f \in \mathcal{B}_{p,q}^s(R)} \mathbb{E}[\|\widehat{f}_n - f\|_{L^r}] \asymp \begin{cases} n^{-\frac{s}{2s+d}}, & \text{if } p \geq r \text{ (dense zone),} \\ \left(\frac{\log n}{n}\right)^{\frac{s-d(1/p-1/r)}{2(s-d(1/p-1/r))+d}}, & \text{if } p < r \text{ (sparse zone).} \end{cases}$$

147 The dense rate is achieved by linear methods (projection, kernel smoothing). The sparse rate requires
148 nonlinear wavelet thresholding; linear estimators are suboptimal by a polynomial factor in $\log n$.

149	4.2	Compression
150	4.3	Flow Matching
151	4.4	Transformers

152 4.5 ASTs

153 **Definition 4.5** (AND–OR Abstract Syntax Tree). Let $\mathcal{X}$ be an input space and $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ an
 154 embedding. An *AND–OR abstract syntax tree (AST)* is defined recursively as:

- 155 • **Atom (Leaf):**

$$T(x) = g(x), \quad g \in \mathcal{G},$$

156 where $\mathcal{G}$ is a class of simple local models (e.g. linear or affine functions).

- 157 • **AND node:**

$$T(x) = \mathbb{I}[C_1(x) \wedge \cdots \wedge C_k(x)] \cdot T'(x),$$

158 where each $C_i(x)$ is a constraint and T' is a subtree.

- 159 • **OR node:**

$$T(x) = \sum_{j=1}^m \mathbb{I}[C_j(x)] \cdot T_j(x),$$

160 where $\{C_j\}_{j=1}^m$ form a partition of $\mathcal{X}$ and T_j are subtrees.

161 The constraints $C(x)$ are typically hyperplane tests of the form

$$C(x) := \langle w, \phi(x) \rangle \geq 0.$$

162 Semantics.

- 163 • AND nodes represent conjunctions of constraints and correspond to intersections of half-
 164 spaces or cross-products of independent factors.
- 165 • OR nodes represent partitions of the input space into disjoint regions.
- 166 • Each root-to-leaf path defines a region

$$\mathcal{X}_\ell = \bigcap_{C \in \text{path}(\ell)} C,$$

167 on which the function reduces to a simple local model.

168 **Example: 1D Piecewise Linear Function** Let

$$f(x) = \begin{cases} a_1x + b_1, & x \leq t, \\ a_2x + b_2, & x > t. \end{cases}$$

169 This function admits the AND–OR representation

$$f(x) = \mathbb{I}[x \leq t] (a_1x + b_1) + \mathbb{I}[x > t] (a_2x + b_2).$$

170 Equivalently, as an AST:

$$\text{OR}\left(\text{AND}(x \leq t, \text{LINEAR}(a_1, b_1)), \text{AND}(x > t, \text{LINEAR}(a_2, b_2))\right).$$

171 **Example: Hierarchical Partition** Let $t_1 < t_2$ and

$$f(x) = \begin{cases} a_1x + b_1, & x \leq t_1, \\ a_2x + b_2, & t_1 < x \leq t_2, \\ a_3x + b_3, & x > t_2. \end{cases}$$

172 Then f can be written as

$$f(x) = \mathbb{I}[x \leq t_2] \left(\mathbb{I}[x \leq t_1] (a_1x + b_1) + \mathbb{I}[t_1 < x \leq t_2] (a_2x + b_2) \right) + \mathbb{I}[x > t_2] (a_3x + b_3).$$

173 This corresponds to a depth–2 AND–OR tree.

174 **Example: ReLU as AND–OR** For $w \in \mathbb{R}^d$,

$$\text{ReLU}(\langle w, x \rangle) = \mathbb{I}[\langle w, x \rangle \geq 0] \langle w, x \rangle + \mathbb{I}[\langle w, x \rangle < 0] \cdot 0.$$

175 Thus a ReLU unit is an OR node with two AND branches. A finite ReLU network corresponds to a
176 (possibly large) AND–OR tree, whose leaves are linear functions and whose OR structure encodes
177 the activation pattern partition.

178

179 [Emile says: What kind of non-trivial theoretical results to validate this construction would be the
180 most insightful? One thing is that for coarse/non-granular functions, it should sort of recover the
181 original manifold hypothesis? What are the theoretical results pertaining to manifold hypothesis in
182 the other papers we've been reading? Can we establish any analogs?]

183 [Emile says: What is the simplest possible way to empirically validate anything that we've constructed
184 (after constructing it)?]

185 A Learning the regularizer

186 Assume, the regularizer depends on some parameter θ . Then learning it is equivalent to minimizing
187 the negative log-likelihood:

$$\min_{\theta} \mathcal{L}(\theta) = \lambda \int_f R_{\theta}(f) d\pi(f) + \log Z(\theta)$$

188 where $Z(\theta) = \int_{f \in X} \exp(-R_{\theta}(f)) \nu(df)$ is a normalizing constant, X is function space, ν base
189 reference measure on X .

190 Or, if we are given the empirical observations of functions $\{f_i\}_{i=1}^t$ then

$$\min_{\theta} \mathcal{L}(\theta) = \frac{\lambda}{t} \sum_i R_{\theta}(f_i) + \log Z(\theta)$$

191 Assuming $R_{\theta}(f) = \langle \theta, \Phi(f) \rangle$ (linear in θ) our problem is convex, yaaaaay! Parameter θ can be
192 here an infinite-dimensional measure or a vector, hence, the minimization problem can be solved by
193 gradient descent on θ .

194 The gradient of the data term is

$$\nabla_{\theta} \left(\frac{1}{T} \sum_{t=1}^T \langle \theta, \Phi(f^{(t)}) \rangle \right) = \frac{1}{T} \sum_{t=1}^T \Phi(f^{(t)}) =: \mathbb{E}_{\text{data}}[\Phi(f)].$$

195 For the partition function term, differentiation under the integral yields

$$\begin{aligned} \nabla_{\theta} \log Z(\theta) &= \frac{1}{Z(\theta)} \nabla_{\theta} Z(\theta) \\ &= \frac{1}{Z(\theta)} \int (-\Phi(f)) \exp(-\langle \theta, \Phi(f) \rangle) \nu(df) \\ &= -\mathbb{E}_{f \sim p_{\theta}}[\Phi(f)]. \end{aligned}$$

196 Combining the two terms, the gradient of the objective is

$$\boxed{\nabla_{\theta} \mathcal{L}(\theta) = \mathbb{E}_{\text{data}}[\Phi(f)] - \mathbb{E}_{f \sim p_{\theta}}[\Phi(f)],}$$

197 A gradient descent step with step size $\eta > 0$ is therefore given by

$$\boxed{\theta_{k+1} = \theta_k - \eta \left(\mathbb{E}_{\text{data}}[\Phi(f)] - \mathbb{E}_{f \sim p_{\theta_k}}[\Phi(f)] \right).}$$

198 If θ is constrained (e.g., $\theta \geq 0$ or θ lies in a simplex), the update is followed by projection onto the
199 feasible set Θ :

$$\theta_{k+1} = \Pi_{\Theta} \left[\theta_k - \eta \left(\mathbb{E}_{\text{data}}[\Phi(f)] - \mathbb{E}_{f \sim p_{\theta_k}}[\Phi(f)] \right) \right].$$

200 At a stationary point θ^* , the optimality condition is the moment-matching identity

$$\mathbb{E}_{\text{data}}[\Phi(f)] = \mathbb{E}_{f \sim p_{\theta^*}}[\Phi(f)].$$

201 $\Phi(f)$ here a statistic extractor on function space.

202 B Examples

203 Note, in all the examples that follow $\Phi(f)$ is convex in f .

Example 1 (Energy regularization).

$$\Phi : L^2(M) \rightarrow \mathbb{R}, \quad \Phi(f) := \|f\|_{L^2(M)}^2.$$

204 The induced regularizer $R(f) = \|f\|_{L^2}^2$ corresponds to ridge/Tikhonov regularization. This choice
205 measures only the magnitude of f and ignores geometric structure.

206 B.1 First-Order Smoothness

Example 2 (Sobolev H^1 regularization).

$$\Phi : H^1(M) \rightarrow \mathbb{R}, \quad \Phi(f) := \int_M \|\nabla f(x)\|^2 dx.$$

207 This feature measures global smoothness. The null space consists of constant functions.

208 **Example 3 (Anisotropic Sobolev regularization).** Let $\{Q_j\}_{j=1}^k$ be fixed symmetric positive
209 semidefinite matrices. Define

$$\Phi : H^1(M) \rightarrow \mathbb{R}^k, \quad \Phi_j(f) := \int_M \langle \nabla f(x), Q_j \nabla f(x) \rangle dx.$$

210 Then

$$R_\theta(f) = \sum_{j=1}^k \theta_j \Phi_j(f)$$

211 corresponds to learning a Riemannian metric.

212 B.2 Total Variation and Sparsity

Example 4 (First-order total variation).

$$\Phi : BV(M) \rightarrow \mathbb{R}_+, \quad \Phi(f) := \|\nabla f\|_{\mathcal{M}}.$$

213 This feature measures the total mass of the gradient as a Radon measure and promotes piecewise-
214 constant functions.

215 B.3 Second-Order and Curvature-Based Statistics

Example 5 (Second-order total variation / splines).

$$\Phi : X \rightarrow \mathbb{R}_+, \quad \Phi(f) := \|\Delta f\|_{\mathcal{M}},$$

216 where X is the space of functions with measure-valued Laplacian.

217 The null space consists of affine functions, and minimizers are adaptive splines.

218 **Example 6 (Anisotropic second-order TV).** Let $\mathcal{S}_+$ denote the cone of symmetric positive
219 semidefinite matrices. Define

$$\Phi : X \rightarrow \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+), \quad \Phi(f)(Q) := \|\Delta_Q f\|_{\mathcal{M}},$$

220 where $\Delta_Q f := \nabla \cdot (Q \nabla f)$.

221 Here:

- 222 • domain: functions with measure-valued anisotropic Laplacian,
- 223 • codomain: nonnegative functions on $\mathcal{S}_+$,
- 224 • regularizer:

$$R_\theta(f) = \int_{\mathcal{S}_+} \|\Delta_Q f\|_{\mathcal{M}} d\theta(Q).$$

225 This promotes sparse, geometry-aligned curvature and induces a Finsler total variation.

226 B.4 Spectral Statistics

227 **Example 7 (Spectral energy features).** Let $\{\psi_k\}$ be eigenfunctions of a Laplacian on M . Define

$$\Phi : L^2(M) \rightarrow \ell^1, \quad \Phi_k(f) := |\langle f, \psi_k \rangle|^2.$$

228 The corresponding regularizer weights energy by frequency and induces a spectral bias.

229 B.5 Measure-Based and Neural Representations

230 **Example 8 (Barron / ReLU representation).** Let

$$f_\mu(x) = \int_{\mathbb{S}^{d-1}} \sigma(\langle w, x \rangle) d\mu(w),$$

231 with μ a finite signed Radon measure. Define

$$\Phi : \mathcal{B} \rightarrow \mathcal{M}_+(\mathbb{S}^{d-1}), \quad \Phi(f_\mu) := |\mu|.$$

232 Here:

- 233 • domain: Barron space $\mathcal{B}$,
- 234 • codomain: finite nonnegative measures,
- 235 • regularizer: $R(f_\mu) = \|\mu\|_{\text{TV}}$.

236 This promotes sparse ridge representations.

237 **Example 9 (Finsler-TV / Radon-Laplacian).** Combining the previous examples,

$$\Phi : \mathcal{B} \rightarrow \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+), \quad \Phi(f_\mu)(Q) := \int \sqrt{w^\top Q w} |\mu|(dw).$$

238 Equivalently,

$$R(f_\mu) = \int_{\mathcal{S}_+} \Phi(f_\mu)(Q) d\theta(Q) = \int F(w) |\mu|(dw), \quad F(w) := \int \sqrt{w^\top Q w} d\theta(Q).$$

239 This is a Finsler total variation norm on kink directions.

240 C Back to the problem

241 Given all distribution over functions minimizing the negative log-likelihood is

$$\min_{\theta} \mathcal{L}(\theta) = \lambda \int_f R_{\theta}(f) d\pi(f) + \log Z(\theta)$$

242 where $Z(\theta) = \int_{f \in X} \exp(-R_{\theta}(f)) \nu(df)$ is a normalizing constant, X is function space, ν base
243 reference measure on X .

244 However, everything we observe is just a discrete sample. Hence, given the empirical observations of
245 functions $\{f_i\}_{i=1}^t$ our minimization looks like

$$\min_{\theta} \mathcal{L}(\theta) = \frac{\lambda}{t} \sum_i R_{\theta}(f_i) + \log Z(\theta)$$

246 Again, **everything** we observe is a discrete sample. Usually, we have access to each function f via
247 its values: $D = \{(x_i, y_i)\}$ where $y_i = f(x_i) + \mathcal{N}(0, \sigma^2)$. Note, that

$$\log p_{\theta}(D_Y | f, D_X) \propto -\frac{1}{\sigma^2} \sum_{(x,y) \in D} \|y - f(x)\|_2^2 = \ell(D, f)$$

248 Hence,

$$-\log p_{\theta}(D_Y | D_X) = -\log \int \exp(-\ell(D, f) - \lambda R_{\theta}(f)) \nu(df) + \log Z(\theta)$$

249 So given datasets $\{D^{(i)}\}_{i=1}^t$ our optimization problem is

$$\min_{\theta} \mathcal{L}(\theta) = -\frac{1}{t} \sum_{i=1}^t \log \int \exp\left(-\ell\left(D^{(i)}, f\right) - \lambda R_{\theta}(f)\right) \nu(df) + \log Z(\theta) \quad (1)$$

250 However, the term $\log \int \exp \dots$ is intractable :(This is why the masking comes into play in practice.

D Masking

D.1 Posterior concentration around the MAP estimator

Fix a parameter θ and a dataset $D = \{(x_j, y_j)\}_{j=1}^n$. Define the energy functional

$$E_\theta(f) := \ell(D, f) + \lambda R_\theta(f),$$

where $\ell(D, f)$ denotes the empirical negative log-likelihood and R_θ is a regularizer.

Assume the following:

1. The dataset D is large (large n) and noise variance σ^2 is small
2. (*Coercivity*) The functional R_θ is coercive on X modulo a finite-dimensional nullspace $\mathcal{N}$, i.e., $\|f\|_X \rightarrow \infty$ with $f \perp \mathcal{N}$ implies $R_\theta(f) \rightarrow \infty$.
3. (*Strict convexity*) The functional $\ell(D, \cdot)$ is strictly convex on $X/\mathcal{N}$.
4. (*Regularity*) The map $f \mapsto E_\theta(f)$ is twice differentiable in a neighborhood of its minimizer.

Under these assumptions, the minimization problem

$$\hat{f}_\theta \in \arg \min_{f \in X} E_\theta(f)$$

admits a unique solution modulo $\mathcal{N}$. Moreover, the posterior distribution

$$p_\theta(f \mid D) \propto \exp(-E_\theta(f))$$

concentrates in probability around $\hat{f}_\theta$ as either $n \rightarrow \infty$ or $\sigma^2 \rightarrow 0$. In particular, for any neighborhood U of $\hat{f}_\theta$ in X ,

$$p_\theta(U \mid D) \rightarrow 1.$$

D.2 Laplace approximation of the marginal likelihood

Let $E_\theta(f) = \ell(D, f) + \lambda R_\theta(f)$ and let $\hat{f}_\theta$ denote its unique minimizer. Define the Hessian operator

$$H_\theta := \nabla^2 E_\theta(\hat{f}_\theta),$$

acting on the tangent space of $X/\mathcal{N}$.

By a second-order Taylor expansion,

$$E_\theta(f) = E_\theta(\hat{f}_\theta) + \frac{1}{2} \langle f - \hat{f}_\theta, H_\theta(f - \hat{f}_\theta) \rangle + o(\|f - \hat{f}_\theta\|_X^2).$$

Substituting this expansion into the marginal likelihood integral yields the Laplace approximation

$$\int_X \exp(-E_\theta(f)) \nu(df) \approx \exp(-E_\theta(\hat{f}_\theta)) \int \exp(-\frac{1}{2} \langle h, H_\theta h \rangle) dh. \quad (2)$$

The remaining integral is Gaussian and evaluates (up to a multiplicative constant) to $(\det H_\theta)^{-1/2}$, where $\det H_\theta$ denotes a suitable regularized determinant. Consequently,

$$\int_X \exp(-\ell(D, f) - \lambda R_\theta(f)) \nu(df) \approx \exp(-\ell(D, \hat{f}_\theta) - \lambda R_\theta(\hat{f}_\theta)) (\det H_\theta)^{-1/2}. \quad (3)$$

D.3 Decomposition of the marginal likelihood objective

Substituting the Laplace approximation (3) into the negative log marginal likelihood yields

$$-\log p_\theta(D_Y \mid D_X) \approx \ell(D, \hat{f}_\theta) + \lambda R_\theta(\hat{f}_\theta) + \frac{1}{2} \log \det H_\theta - \log Z(\theta), \quad (4)$$

up to additive constants independent of θ .

This expression decomposes the marginal likelihood into four interpretable components:

- 276 1. *Data-fit term* $\ell(D, \hat{f}_\theta)$, measuring how well the MAP estimator explains the observations;
- 277 2. *Regularization cost* $\lambda R_\theta(\hat{f}_\theta)$, encoding alignment between the learned function and the
- 278 inductive bias imposed by θ ;
- 279 3. *Local complexity (Occam) penalty* $\frac{1}{2} \log \det H_\theta$, which penalizes flat or unstable directions
- 280 around the MAP solution;
- 281 4. *Global complexity penalty* $-\log Z(\theta)$, measuring the overall volume of functions favored
- 282 by the prior.

283 Together, these terms characterize the tradeoff between data fit, regularization, and model complexity
 284 in the marginal likelihood objective.

285 E Back to the optimization

286 Given the collection of datasets $\mathcal{D} = \left\{ \left(D_{\text{train}}^{(i)}, D_{\text{test}}^{(i)} \right) \right\}_{i=1}^t$ Substituting this into Equation (1) we
 287 have

$$\min_{\theta} \mathcal{L}(\theta) = \frac{1}{t} \sum_{i=1}^t \ell \left(D_{\text{test}}^{(i)}, \hat{f}_\theta^{(i)} \right) + \lambda R_\theta \left(\hat{f}_\theta^{(i)} \right) + \frac{1}{2} \log \det H_\theta - \log Z(\theta)$$

where $\hat{f}_\theta^{(i)} = \operatorname{argmin}_f \ell \left(D_{\text{train}}^{(i)}, f \right) + \lambda R_\theta(f)$

288 This is bilever optimization. Even though it lost the convexity, it is tractable and can be solved in
 289 practice by alternating gradient descent. I claim that gradient descent of the outer problem = training
 290 a transformer, gradient descent of the inner problem is a transformer forward pass.

291 F Inner problem

292 For a dataset $D = \{(x_i, y_i)\}$ we have

$$\hat{f}_\theta = \operatorname{argmin}_f \ell(D, f) + \lambda R_\theta(f) = \operatorname{argmin}_f \|f(x_i) - y_i\|_2^2 + \lambda \langle \theta, \Phi(f) \rangle$$

293 Note, that if Φ is convex in f then this can be solved by subgradient descent on f :

$$f^{(t+1)} = f^{(t)} - \eta \left[2 \sum_i (f^{(t)}(x_i) - y_i) \delta_{x_i} + \lambda g^{(t)} \right]$$

294 where $g^{(t)} \in \partial_f R_\theta(f^{(t)})$ is a subgradient of the regularizer.

295 F.1 How transformer forward pass can implement a (sub)gradient descent

296 Now lets go through some of the examples in Appendix B.3 to see how transformer-like networks
 297 can implement a (sub)gradient g descent for the inner problem.

Energy regularization

$$\Phi : L^2(M) \rightarrow \mathbb{R}, \quad \Phi(f) := \|f\|_{L^2(M)}^2 \quad R_\theta(f) = \theta \|f\|_{L^2(M)}^2 \quad g = 2f$$

298 **Anisotropic Sobolev Regularization** Parameter θ is a measure over PSD matrices.

$$\Phi(f)(Q) := \|\nabla f(x)\|_Q^2 \quad R_\theta(f) = \int_{Q \in \mathbf{S}_+} \|\nabla f(x)\|_Q^2 d\theta(Q) \quad g = - \int_{Q \in \mathbf{S}_+} \Delta_Q f d\theta(Q)$$

Anisotropic second-order TV

$$\Phi : X \rightarrow \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+), \quad \Phi(f)(Q) := \|\Delta_Q f\|_{\mathcal{M}}, \quad g = \int_{\mathcal{S}_+} \Delta_Q \operatorname{sign}(\Delta_Q f) d\theta(Q)$$

299 where $\Delta_Q f := \nabla \cdot (Q \nabla f)$.

300 As the functions are defined on a "flat" domain $\mathcal{M} \subset R^d$, the discrete version of Δ_Q looks like:

$$L \in R^{n \times n} : L_{ij} = \exp \left(\frac{-\|x_i - x_j\|_{Q^{-1}}^2}{\epsilon} \right)$$

301 If our domain is curved manifold $\mathcal{M}$ we can W.L.O.G. (by Takahashi's thm hehe) assume that it's
 302 $\mathcal{M} \in S^{d-1}$, hence, the discrete version of Δ_Q will require projection $P_i = I - x_i x_i^\top$:

$$\begin{aligned} L \in R^{n \times n} : L_{ij} &= \exp \left(\frac{-(x_i - x_j)^\top P_i Q^{-1} P_i (x_i - x_j)}{\epsilon} \right) \\ &= \exp \left(-\frac{(x_j - (x_i^\top x_j) x_i)^\top Q^{-1} (x_j - (x_i^\top x_j) x_i)}{\epsilon} \right) \end{aligned}$$

303 Note, that all the exp terms are kind of similar to the original transformer softmax terms. I think,
 304 softmax should be replaced by those. The final idea is that a transformish NN with infinite amount of
 305 heads approximates the distribution over Q .

306 TODO: move this

307 **Theorem F.1** (Stability from coercive Φ -regularization). *Let $(X, \|\cdot\|_X)$ be a reflexive Banach space*
 308 *of functions on a domain M , and let*

$$R_\theta(f) := \langle \theta, \Phi(f) \rangle \in [0, \infty]$$

309 *where $\Phi : X \rightarrow V$ is a measurable map into a real locally convex space V and $\theta \in V^*$ is such that*
 310 *R_θ is a proper, convex, lower semicontinuous functional on X . Assume the following:*

311 1. **Coercivity (modulo a nullspace).** *There exists a finite-dimensional subspace $\mathcal{N} \subset X$ and a*
 312 *constant $c > 0$ such that for all $f \in X$,*

$$R_\theta(f) \geq c \|f - \Pi_{\mathcal{N}} f\|_X - c,$$

313 *where $\Pi_{\mathcal{N}}$ denotes a fixed linear projection onto $\mathcal{N}$. In particular, the sublevel sets $\{f :$
 314 $R_\theta(f) \leq C\}$ are relatively compact in $X/\mathcal{N}$.*

315 2. **Strong convexity of the empirical loss.** *For each dataset $D = \{(x_i, y_i)\}_{i=1}^n$, define*

$$\mathcal{L}_D(f) := \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i),$$

316 *and assume $\mathcal{L}_D$ is α -strongly convex on $X/\mathcal{N}$ with respect to $\|\cdot\|_X$, uniformly over all*
 317 *datasets D (for some $\alpha > 0$).*

318 3. **Lipschitz dependence on function values.** *Assume $\ell(\cdot, y)$ is L -Lipschitz in its first argument*
 319 *for all y , and that point-evaluation is continuous on X in the sense that there exists $\kappa > 0$*
 320 *such that for all $x \in M$ and all $f, g \in X$,*

$$|f(x) - g(x)| \leq \kappa \|f - g\|_X.$$

321 *Observed data $y_i = f^*(x_i) + \mathcal{N}(0, \sigma^2)$*

$$\hat{f} = \argmin_f \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i) + \lambda R(f) \Rightarrow \|f^* - \hat{f}\|_X \leq C(n, \sigma, R, \lambda)$$

322 **Proposition F.2** (Bilevel learning of θ as a MAP surrogate). *Let $R_\theta(f) = \langle \theta, \Phi(f) \rangle$ and define the*
 323 *inner estimator*

$$\hat{f}_\theta(D) \in \argmin_{f \in X} \left\{ \mathcal{L}_D(f) + \lambda R_\theta(f) \right\}.$$

324 *Consider the bilevel objective based on masking (or train/validation splits),*

$$\min_{\theta} \mathbb{E}_{\text{mask}} \left[\mathcal{L}_{\text{val}}(\hat{f}_\theta(D_{\text{train}})) \right] + \Omega(\theta).$$

325 *Then bilevel optimization selects θ so as to minimize the predictive risk of the MAP estimator induced*
 326 *by the Gibbs prior*

$$p_\theta(df) \propto \exp(-\lambda R_\theta(f)) \nu(df).$$

327 *In contrast, maximum-likelihood learning of θ for the induced Gibbs model includes an additional*
 328 *global complexity term $\log Z(\theta)$, which is generally not captured by the bilevel objective unless*
 329 *explicitly approximated (e.g., via Laplace/Occam corrections).*

330 **Proposition F.3** (Bilevel learning of θ as a MAP surrogate). *Let $R_\theta(f) = \langle \theta, \Phi(f) \rangle$ and define the*
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